# Pattern Recognition

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All lecture notes, etc. will be published in the course on LeOn:

**https://leon.pw.edu.pl/**

Lectures: Tuesday 8.15-10.00 (room 170)

(Semester calendar: https://studia3.elka.pw.edu.pl/PL/26L/-/cal/)

# Grading – 103D-CBxxx-MSP-EPART

Overall 64 points are from the theoretical part only:

Theoretical part only – 64 points (test I: max. 31 and test II max. 33 points). Tests will be solved in person in room 170; some examples of test tasks will be published in the second half of March on LeOn.
Important dates:
**April 14th**: test I (lectures 1-6)
**June 9th**: test II (lectures 8-13)
To pass theoretical part you have to achieve at least 32 points.

Grades are mapped linearly from 32..64 range.
During ADAC workshop there will be (probably) five points to achieve – I hope that it will be clear soon.

# Bibliography


Duda R.O., Hart P.E., Stork D.G., *Pattern Classification*, Wiley-Interscience, 2000

Bishop C.M., *Pattern Recognition and Machine Learning*, Springer 2006

Goodfellow I., Bengio Y., Courville A., *Deep Learning*, MIT Press, 2016 (http://www.deeplearningbook.org/)

Jain A. K., *Fundamentals of Digital Image Processing*, Prentice-Hall International Editions, Engelwood Hills, 1989

Press W. H., *Numerical Recipes in C*, Cambridge University Press, Cambridge 1992 (http://www.nrbook.com)

# Learning outcomes

Knowledge:

1. Student knows basic pattern classification methods
2. Student knows initial data analysis and clustering methods
3. Student knows basic methods of designing classifiers' committees

Skills:

1. Student can analyze training set, design simple classifier and assess its quality
2. Basing on the analysis of the training set student is able to select proper classification method and determine its parameters
3. Student is able to assess a pattern classification solution and propose its enhancements

# Pattern recognition

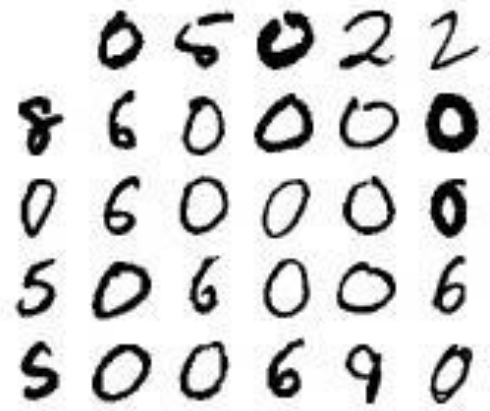
= *pol.* Rozpoznawanie obrazów

**Pattern**

1. a repeated decorative design, e.g. on fabric or china
2. a natural or chance arrangement or sequence
   *a frost pattern*
   *the pattern of events*
3. a design, model, or set of instructions for making things
   *a dress pattern*
4. a model for making a mould into which molten metal is poured to form a casting
5. a form or model proposed for imitation; an example
6. a specimen or sample, e.g. of wallpaper

# Patterns

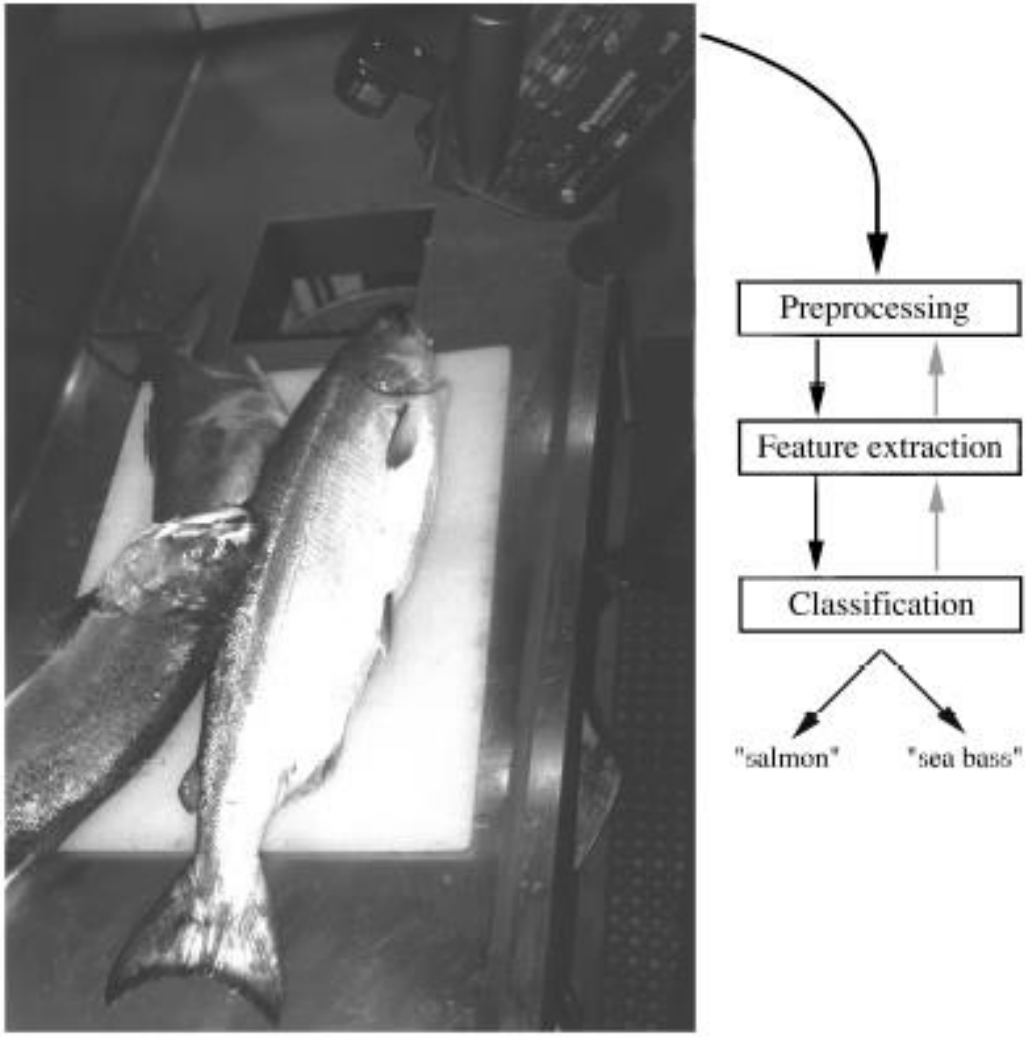
```
0.11586,0.86382,4.2998,0.88029
0.031722,-0.97325,-0.18586,1.154
0.049696,0.39263,-6.17,0.47397
-0.017119,0.096331,-4.3949,0.68433
0.22962,-0.085538,-1.7988,-1.2181
-0.11064,-1.0803,2.1374,0.21424
-0.13247,-1.4722,4.8275,0.84576
0.1299,-0.12251,-0.66075,-1.1466
0.096487,-0.40473,0.91897,1.052
0.036473,-0.33991,3.959,1.2906
```



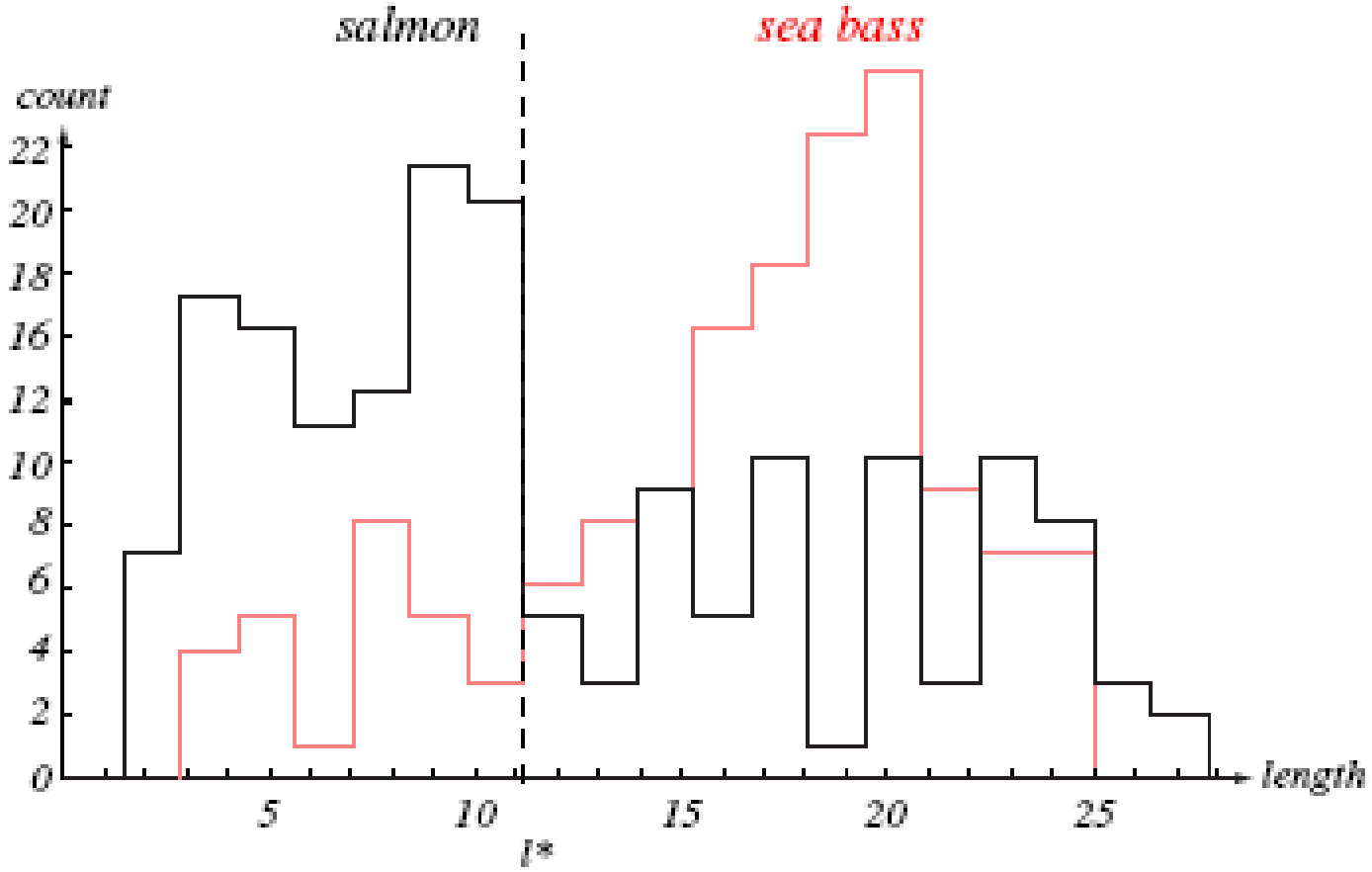




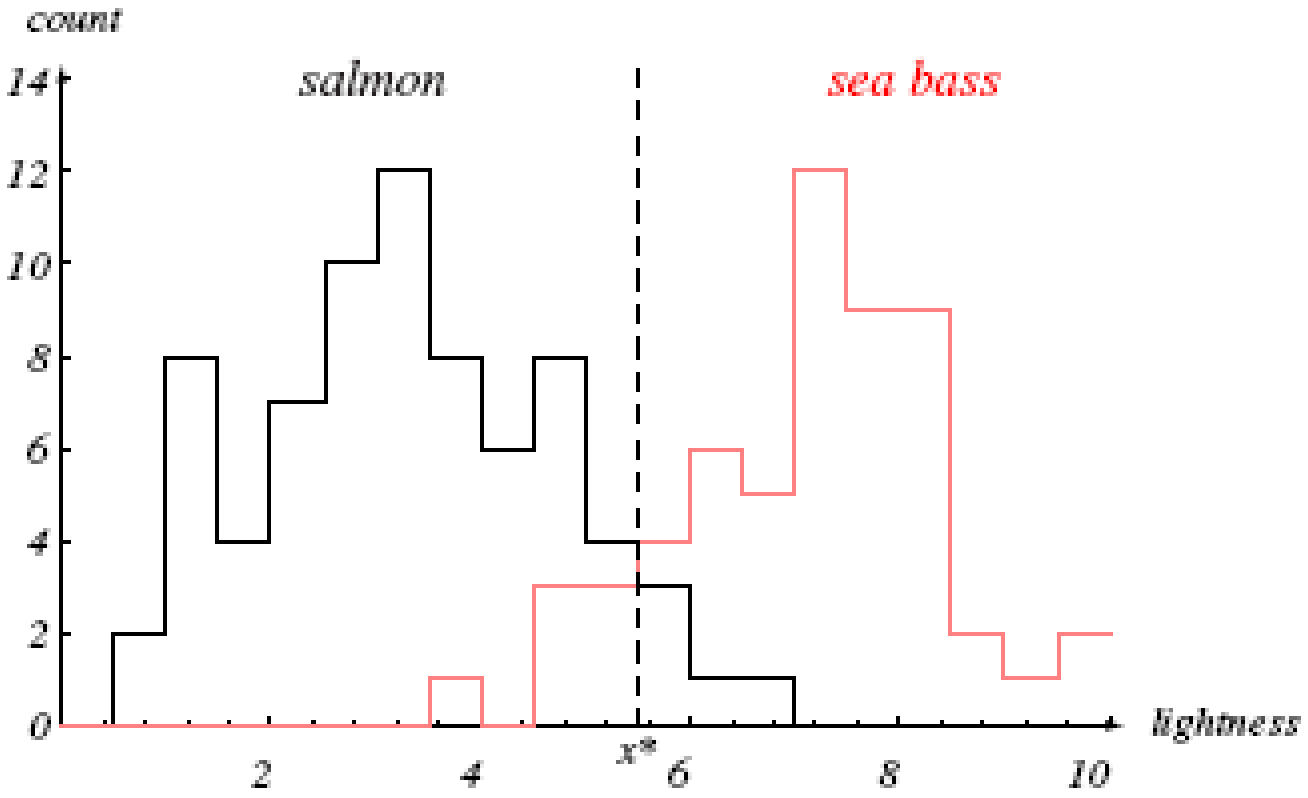
# Example – fish sorting



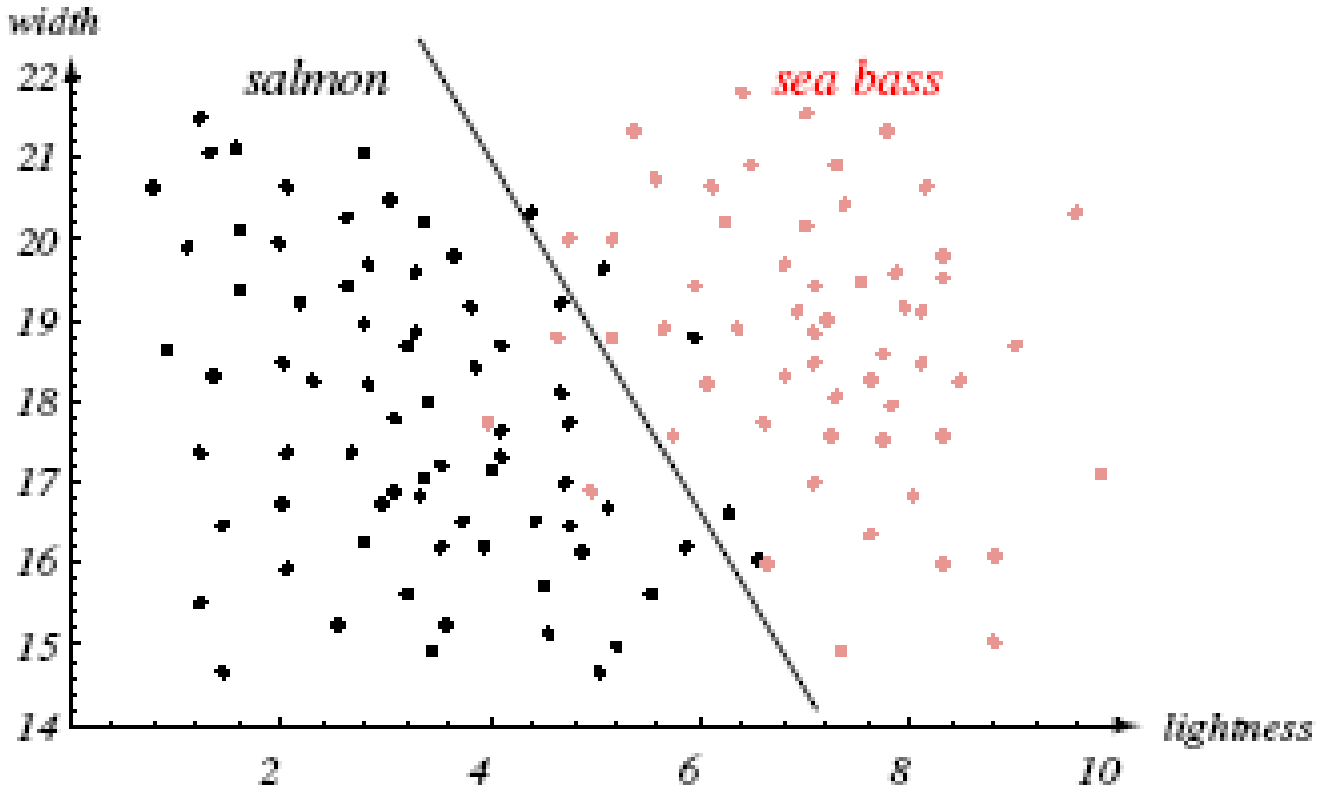
# Feature selection



# Feature selection

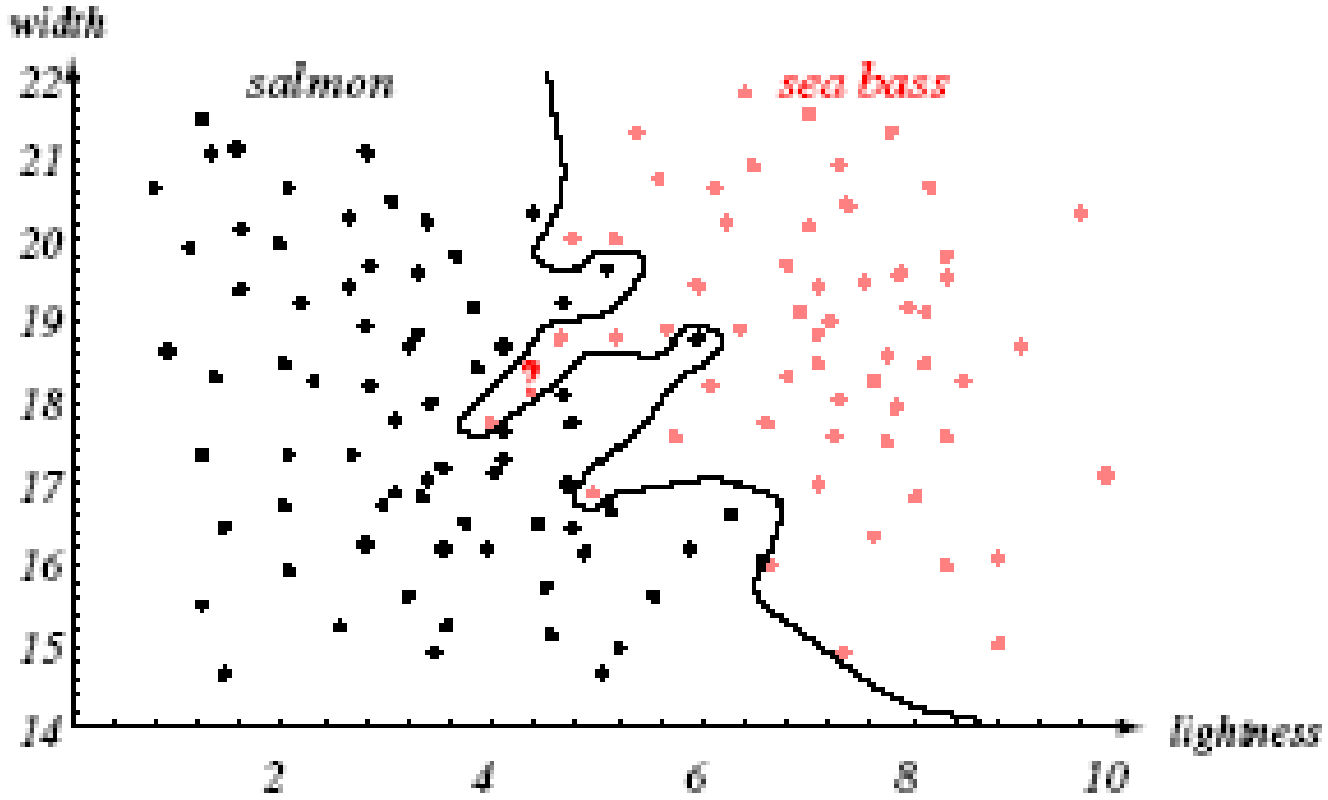


# Feature selection



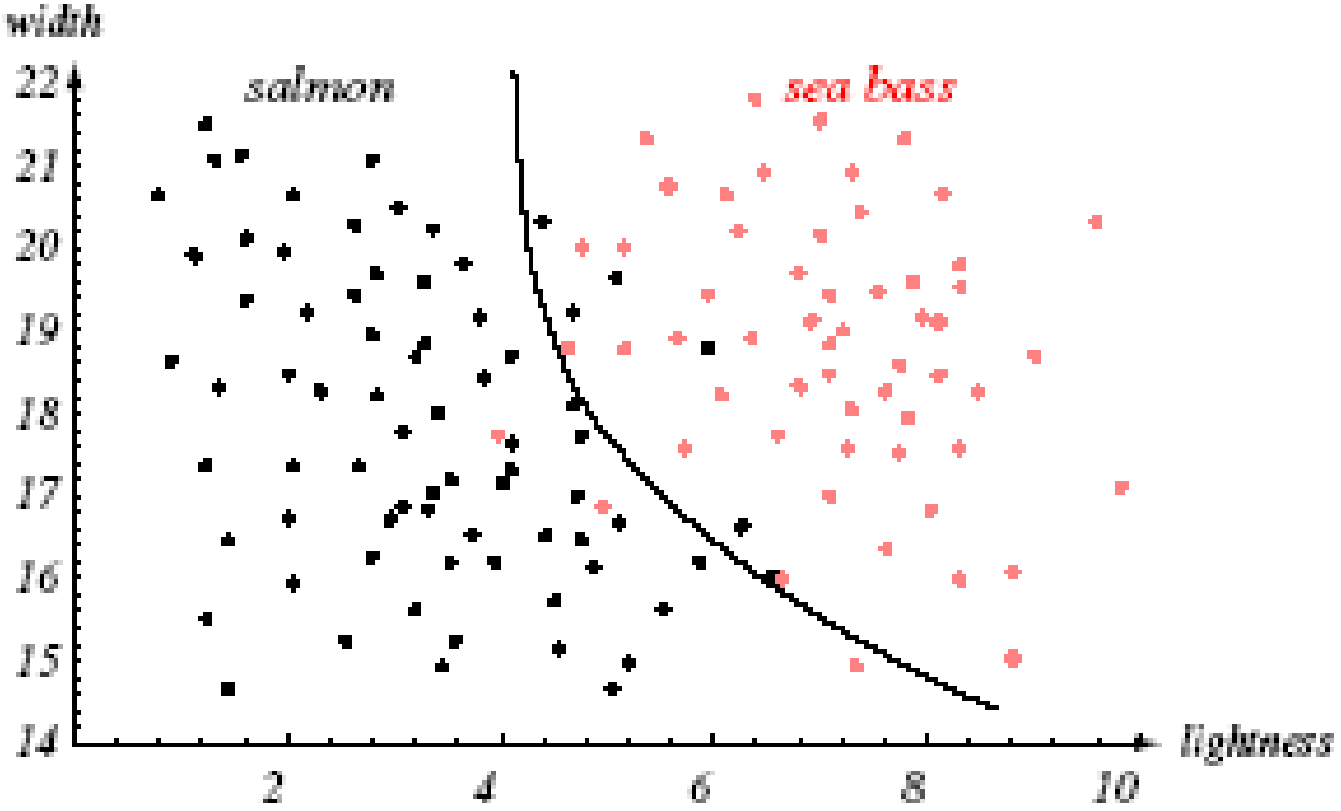

We can use the simplest decision boundary i.e. a line in two dimensional case.

# Classifier training



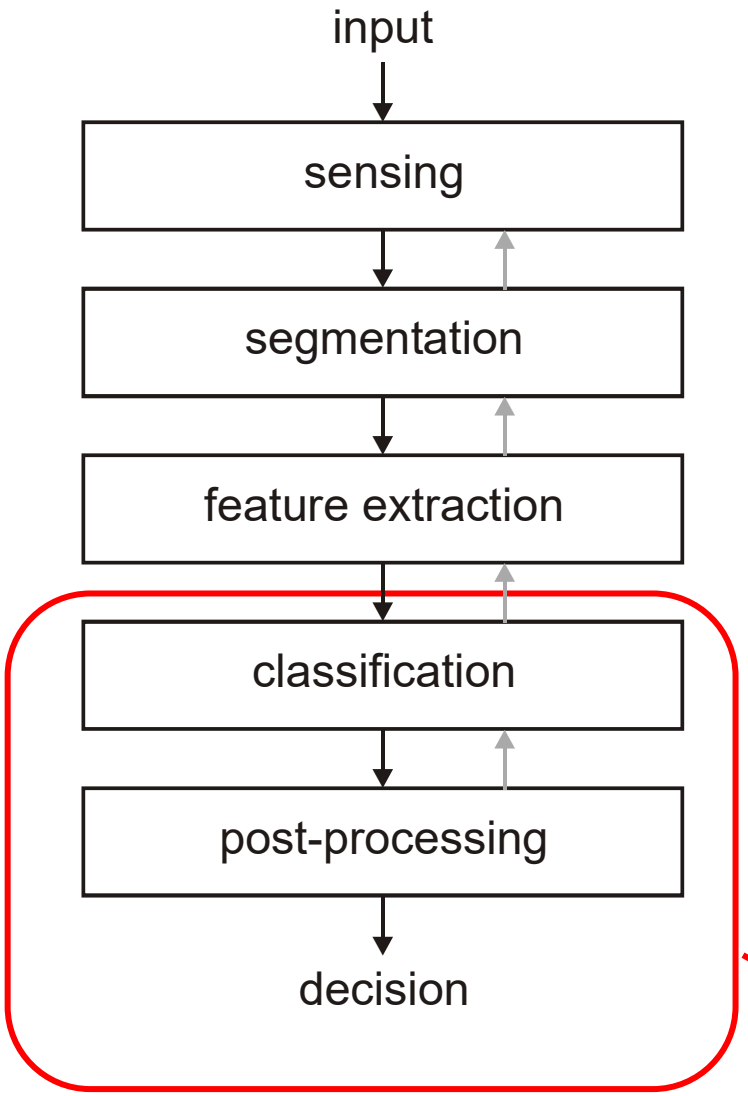

Decision boundary visible on the diagram is too complicated – this classifier will perform poorly on novel data. This phenomenon is called *overfitting of classifier*.

# Classifier training




Although this classifier is not perfect on the training set, we can hope that its performance on real data will be better than that of the previous classifier.

# Pattern recognition system




Very important step, often decides about recognition quality.

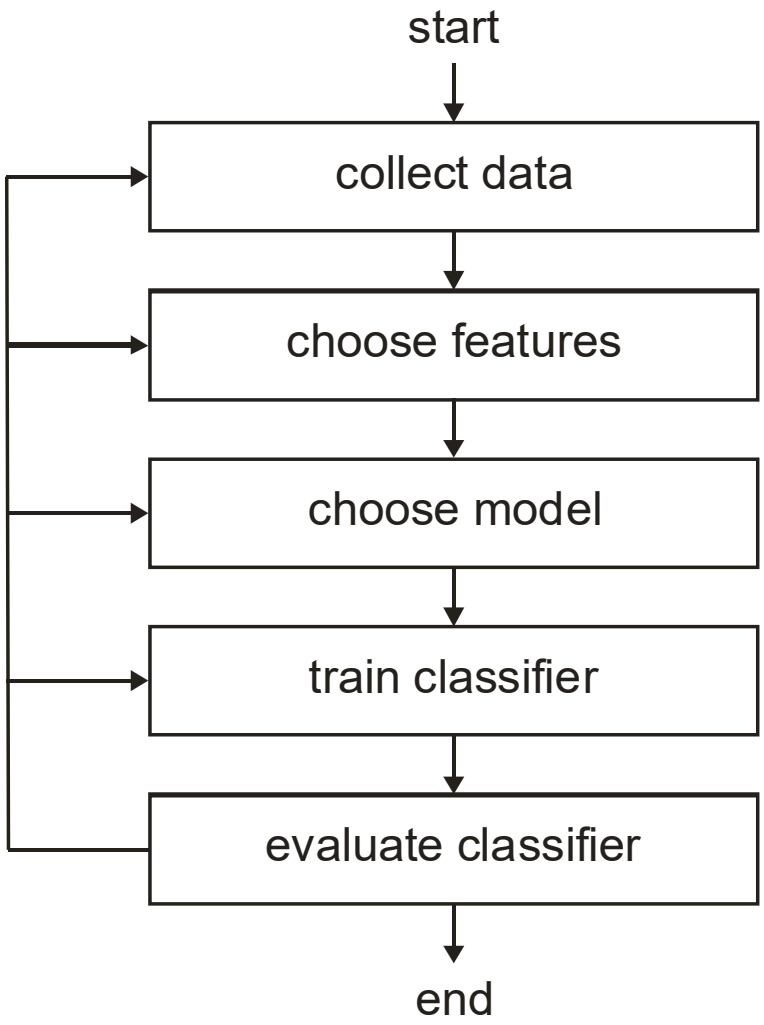
Feature invariance!

Missing features,
use of context information

Use of context information,
risk estimation

EPART interest domain

# Design cycle




Prior domain knowledge.

Model invariance with respect to the expected distortions.

Possibility to fit too much to the training data (*overfitting*).

# Subject matter

1. Introduction
2. Nearest neighbourhood methods
3. Optimal Bayes classification
4. Linear classification
5. Dimensionality reduction
6. Clustering
7. Test I

# Subject matter

8. Neural networks I
9. Neural networks II
10. Markov models
    Exact string matching
11. Approximate string matching
12. Decision trees
13. Recognition quality enhancement I
14. Test II
15. Recognition quality enhancement II

# Top 10 Algorithms in Data Mining discussed on EPART

+ C4.5 and beyond (decision trees)
+ The k-means algorithm
+ Support vector machines
- The Apriori algorithm
- The EM algorithm
- PageRank
+ AdaBoost
+ kNN – k-nearest neighbour classification
+ Naïve Bayes
+ CART

Wu, Xindong et al, Top 10 algorithms in data mining, Journal of Knowledge and Information Systems, Volume 14 Issue 1, December 2007, Pages 1-37